

World Engineering Education Forum 2026 · Cartagena, Colombia

Centralizing the Student Lens

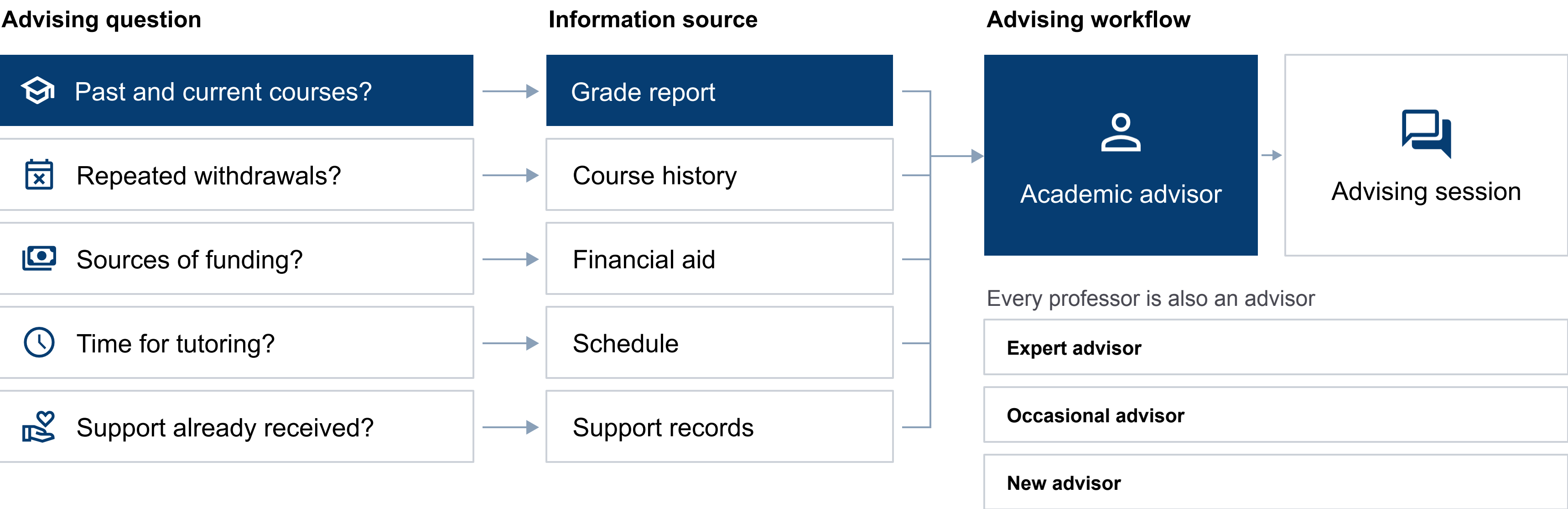
Design and Evaluation of a Learning Analytics Dashboard for Academic Advising in Higher Education

Santiago Martínez Novoa · Federico Melo Barrero · Mariana Ruiz Giraldo · Sofía Velásquez Marín · Nicolás Carvajal Chaves · Marcela Hernández Hoyos · Juan Pablo Reyes

Systems and Computing Engineering Department · EdTech Node Universidad de los Andes · Bogotá D.C., Colombia

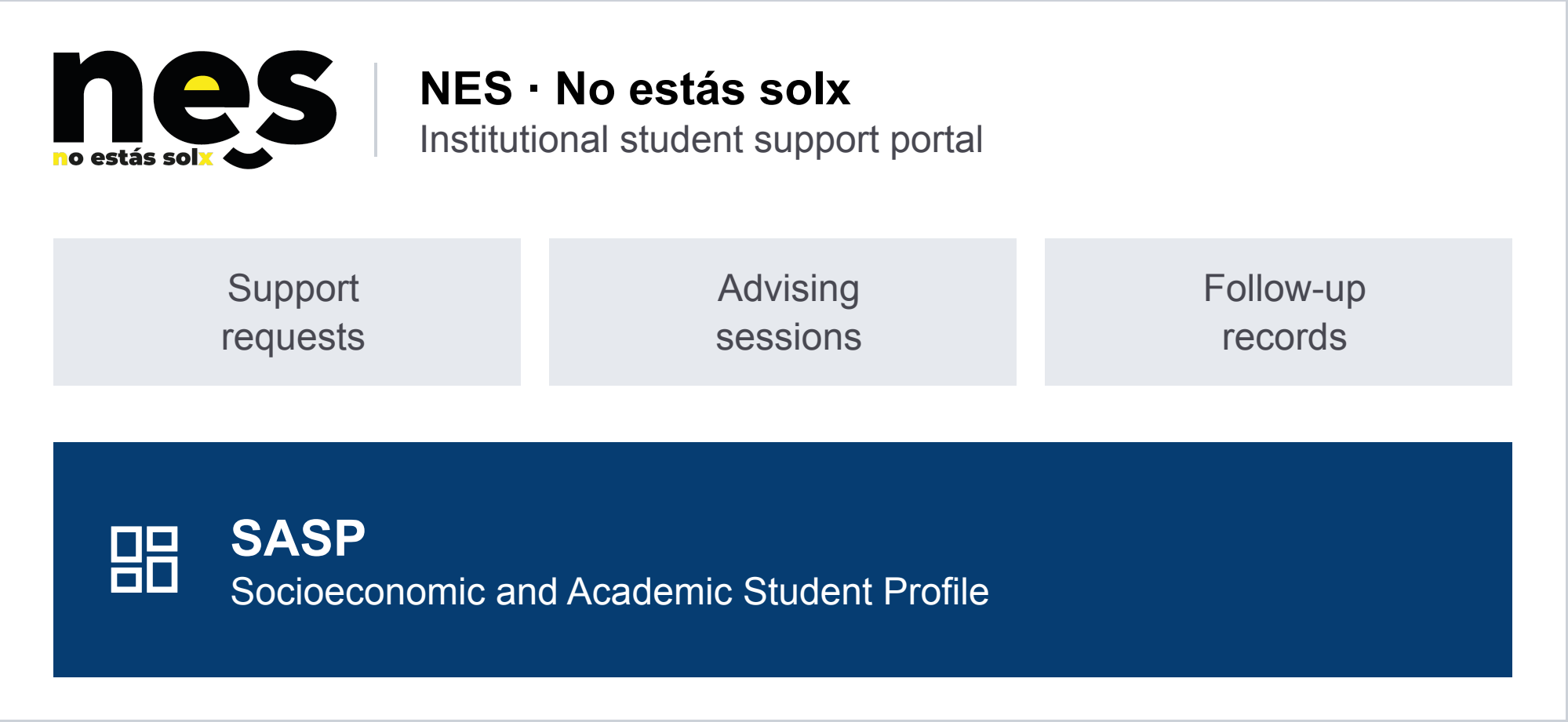
Student information is fragmented

What is the student's current situation?



The advisor assembles the answers before deciding how to help.

SASP within the NES support portal



Weekly co-design

With the Vice Dean's Office for Student Affairs in Engineering



January 2024

Co-design begins



Working versions

Released and revised around user-reported needs

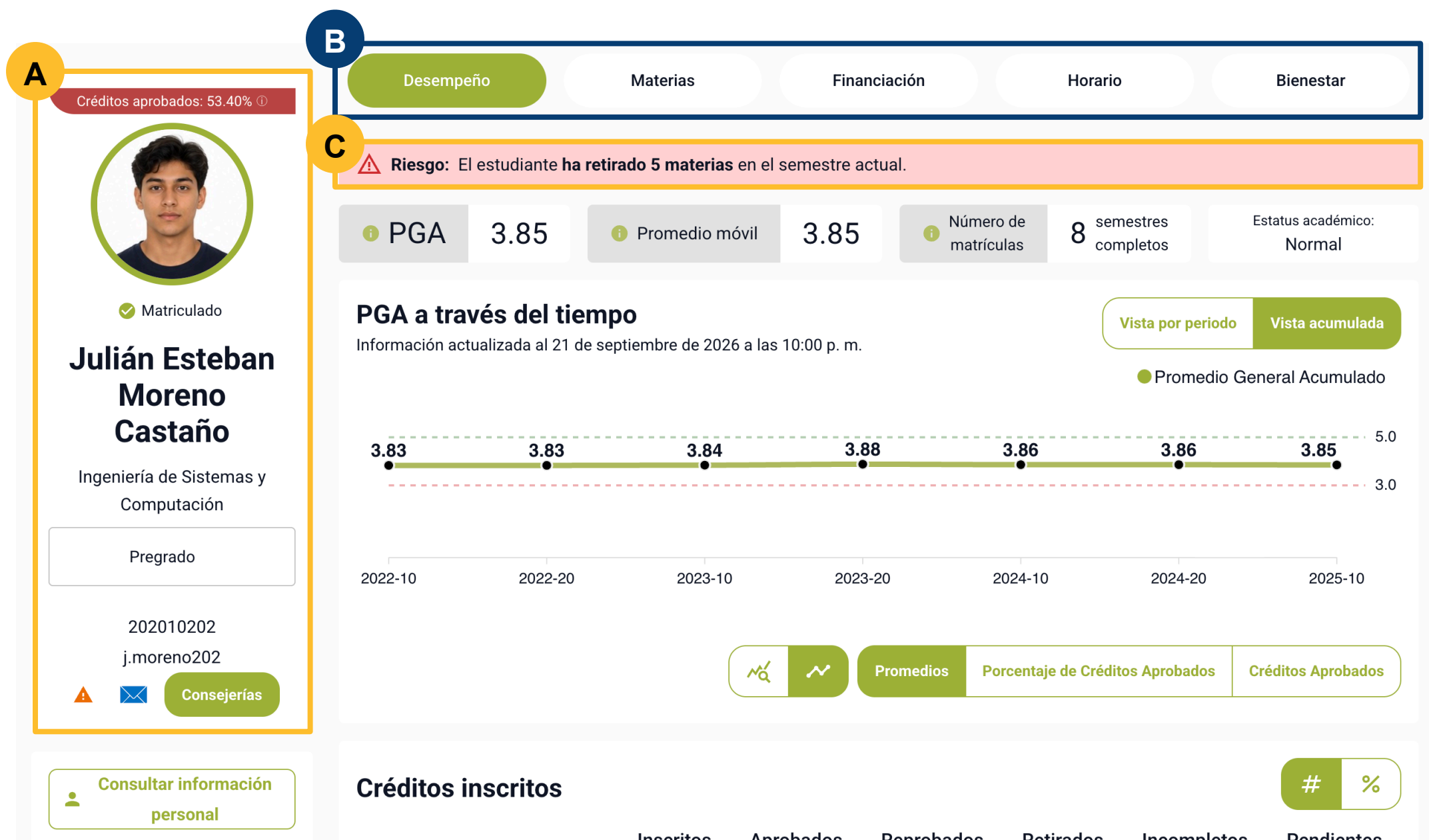


January 2025

End of the co-design period

Next: an advisor preparing to meet a student struggling with mathematics.

Step 1 · The advisor opens the student's profile



Overview first, then a particular concern.

Step 2 · The advisor reads the risk alerts

In the profile · Performance tab

 **Riesgo:** El estudiante está en **tercer semestre o superior** y tiene un **PGA entre 3.25 y 3.50**.

 **Riesgo:** El estudiante tiene un **promedio móvil** inferior a **3.50**.

 **Riesgo:** El estudiante está **bloqueado** por la materia **MATE-1105**, la cual no ha aprobado en **11 intentos**.

 **Riesgo:** El estudiante está **bloqueado** por la materia **FISI-1018**, la cual no ha aprobado en **10 intentos**.

 **Riesgo:** El estudiante está **bloqueado** por la materia **MATE-1214**, la cual no ha aprobado en **10 intentos**.

 **Riesgo:** El estudiante está **bloqueado** por la materia **ECON-2106**, la cual no ha aprobado en **8 intentos**.

 **Riesgo:** El estudiante está **bloqueado** por la materia **ISIS-1107**, la cual no ha aprobado en **5 intentos**.

 **Riesgo:** El estudiante está **bloqueado** por la materia **LENG-1159**, la cual no ha aprobado en **3 intentos**.

By email · to the assigned advisor





Universidad de
los Andes
Colombia

Decanatura
de Estudiantes

No
estás
solx



Cordial saludo profesor/a William,

Con las notas definitivas del semestre ya publicadas, hemos identificado que algunos de sus estudiantes aconsejados tuvieron un promedio semestral inferior a 3.0, lo que implica la pérdida del semestre. Esta situación requiere su intervención para prevenir la deserción y apoyar la recuperación académica del estudiante.

- 1

Plain-language conditions
- 2

Concerns visible beyond the grade
- 3

Same criteria, sent by email

Step 3 · The advisor picks the level and filters by subject

Level selector

Créditos aprobados: 95.90%



Matriculado

Laura María Villamil Rojas

Doctorado en Ingeniería

Pregrado

Pregrado

Magíster

Doctorado

Courses tab · subject-area filter set to MATE

Desempeño

Materias

Financiación

Horario

Bienestar

Filtrar por periodo:

Todos

Filtrar por código de área:

MATE

Filtrar por código de materia:

Todos

Filtrar por estado:

Todos

Mostrar monitorías

Mostrar elementos de 0 créditos

Mostrar deportes

Periodo	Código	Nombre del curso	Créditos	Nota 30%	Nota final	Estado
2026-20	MATE-1105	Algebra Lineal 1	3			
2026-20	MATE-1214	Cálculo Integral-ecuac.Diferen	3			
2026-10	MATE-1102	Matemática Estructural	3	0.00	Retirado	
2025-20	MATE-1105	Algebra Lineal 1	3	1.34	Retirado	
2025-10	MATE-1105	Algebra Lineal 1	3	2.71	1.50	
2025-10	MATE-1214	Cálculo Integral-ecuac.Diferen	3	0.00	1.50	

Levels kept separate

Metrics, alerts and support records update together.

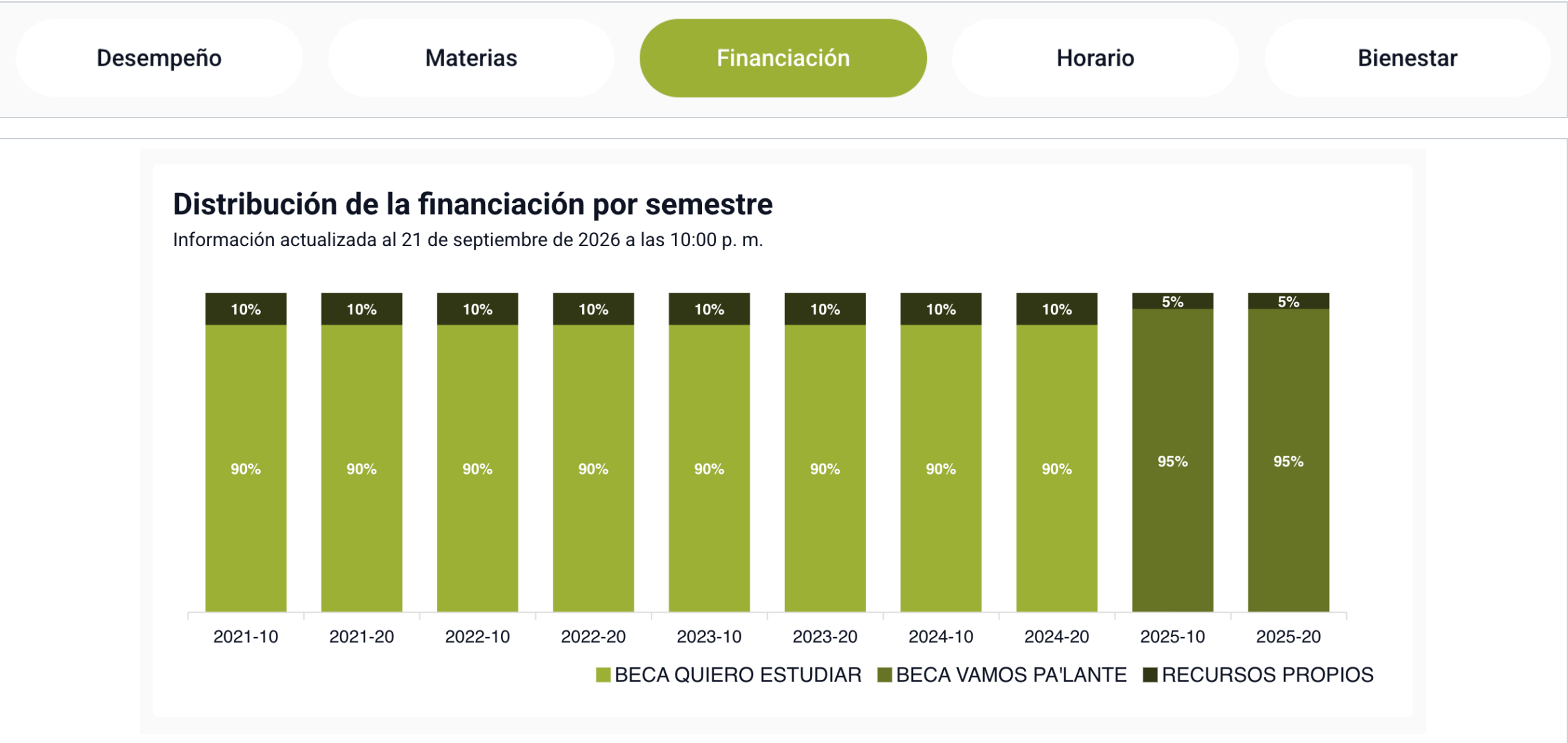
Withdrawals and repeated attempts

Shared with Performance

Credit summary and trend chart, one filter.

Step 4 · The coordinator checks funding

Funding tab · visible to academic coordinators



How has this student financed their studies?

- Personal funds
- Loans
- Scholarships
- A combination, term by term

Repeating courses: extra semesters, and will support continue?

Step 5 · The advisor reviews support already received

Support sources

 Academic counseling

 Dean of Students

 Tutoring records

One timeline · Wellness tab

DesempeñoMateriasFinanciaciónHorarioBienestar

✓ Matriculado

✗ Pénsam anterior (no reformado)

✗ No puede extracreditarse

Máximo de créditos para inscripción18

ConsejeríasDecanatura de EstudiantesRefuerzo académicoHistórico

Histórico

Accede al registro y seguimiento de las sesiones que ha tenido el estudiante hasta el momento.

Mostrando 3 de 3 consejerías

1. Sept. 2026

JM

Juliana Martínez Gómez

Presencial

Consejería

Ver detalle →

12. Aug. 2026

Centro de Diversidad

Consejería Presencial (CD)

Finalizada

Ver detalle →

11. Aug. 2026

Cupi Taller - Tutoría de Introducción a la Programación

Tutor presencial

Finalizada

Ver detalle →

In the mathematics example

No tutoring recorded

Ask about prior help and referrals.

Difficulties persist despite support

Review what was tried.

From a concern to its context, before the conversation.

SASP · Universidad de los Andes

8

Survey and usage data

Voluntary survey

117 responses Compared SASP with previous tools

↓ Also used SASP during a counseling session?

56 session users

Questions about the session

Usage records

9 months

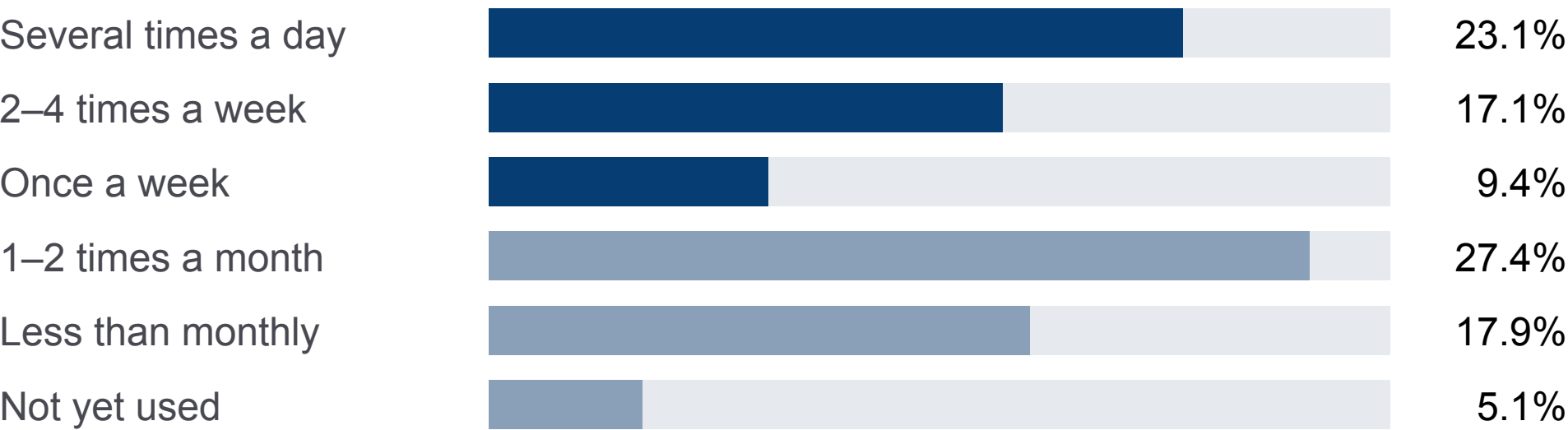
Navigation events from production use

What advisors valued, and whether they kept coming back.

Reported use and return visits

Survey · 117 respondents

94.9% had used SASP



At least weekly · 49.6%

Bars share a 0–30% scale

Use continued beyond the first visit.

Usage records · 9 months

72.7%

returning users



What advisors valued

Survey · 117 respondents

87.2%



Data centralization facilitates my work
Highest-rated of eight items

Session users · n = 56

91.1%



Consider SASP useful during a session
Q12 · mean 4.46 of 5

Session users · n = 56

73.2%



Enough information to be the only tool for the session
Q17

Where we still need to improve

Enables preventive actions



61.5% Performance relative to peers



55.4%

Agreement · Q7 all respondents, Q15 session users

Enough context for the conversation.

How advisors described using SASP

1 Noticing a concern



Email alerts guide outreach



Review at the start of each term

Enrolled and approved credits

2 Looking into the history



Good GPA, little degree progress



One course failed seven times

Three risk alerts, one block

3 Considering follow-up



Acting before students ask
Even when they do not respond

Cases advisors could not detect before.

Limitations and future work

What the study has not yet shown

 **Effects on students' academic outcomes**

 **Voluntary participation**
More engaged advisors may be overrepresented.

 **One university**

Next steps

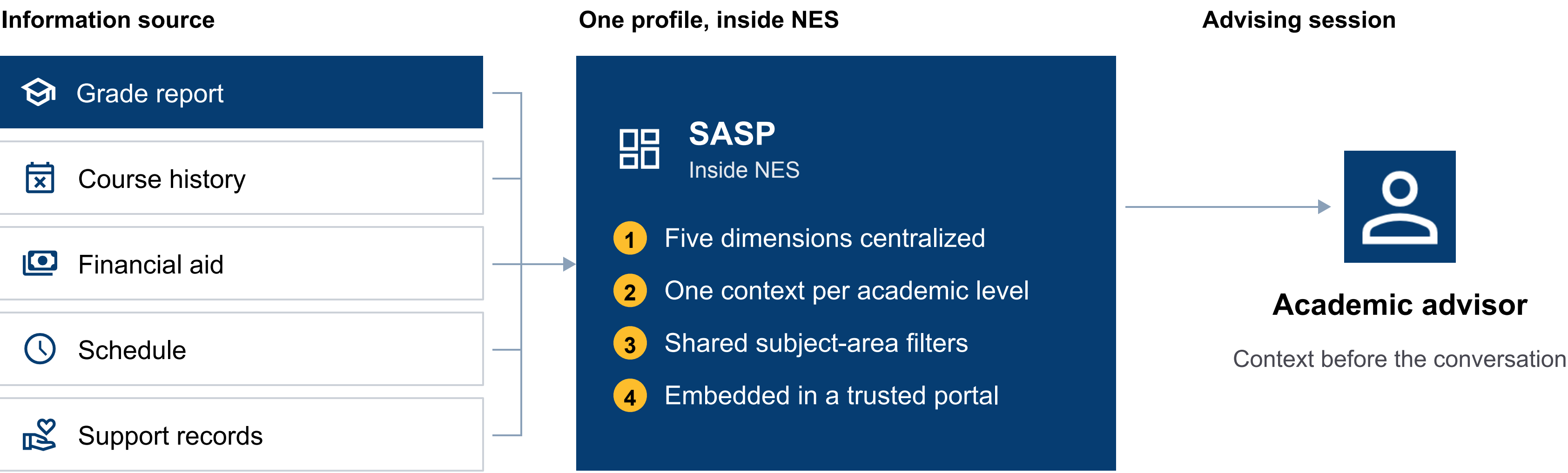
 **Hear from students and follow outcomes over time**

 **Improve preventive action and peer comparison**

 **Test the academic-level selector with users**

 The effect on students is the next question.

Student information, brought together



Context in everyday advising, in a dashboard advisors return to.

Thank you

Explore the current version of SASP at our stand.

Santiago Martínez Novoa
s.martinezn@uniandes.edu.co